

Question number	Answer	Mark
7(a)	The only correct answer is A (position R, position P) <i>B is incorrect because P will have a carboxyl group and R will have an amino group (Q will have a carboxyl group)</i> <i>C is incorrect because P will have a carboxyl group and R will have an amino group (Q will have a carboxyl group)</i> <i>D is incorrect because P will have a carboxyl group and R will have an amino group (Q will have a carboxyl group)</i>	(1)

Question number	Answer	Additional guidance	Mark
7(b)(i)	39 / 39.3 / 39.29 (1)		(1)

Question number	Answer	Additional guidance	Mark
7(b)(ii)	0.13 : 1 / 0.1 : 1 / 0.13 / 0.1 (1)	ACCEPT 1 : 7.5 / 1 : 8	(1)

Question number	Answer	Additional guidance	Mark
7(c)	A description that includes the following points: - hydrolysis / using enzymes (1) - of the peptide bonds between P and the connecting chain <u>and</u> R and the connecting chain (1)	ACCEPT protease ACCEPT of peptide bonds between A chain and connecting region <u>and</u> B chain and connecting region	(2)

Question number	Answer	Additional guidance	Mark
7(d)	An answer that includes the following points: Similarities : • alpha glucose (1) • contain C, H and O (only) (1) Differences : • glucose is a monosaccharide and glycogen is a polysaccharide (1) • glycogen has (1-4 / 1-6) glycosidic bonds but glucose does not (1)	DO NOT PIECE TOGETHER	(3)

Question number	Answer	Mark
*7(e)	Indicative content: Healthy rats given water • levels of glucose {fell slightly / maintained} (D) • because the healthy rats were producing insulin (E) • so were controlling the blood glucose levels naturally (E) Healthy rats given cinnamon • levels of glucose very slightly lower (D) • as cinnamon was adding to the effect of insulin (CE) - OR • levels of glucose very similar (D) • as cinnamon had no (added) effect in healthy rats (CE) Diabetic rats given water • levels of glucose rose (D) • levels of glucose were higher than healthy rats (D) • because diabetic rats had no insulin to control their blood glucose levels (E) Diabetic rats given cinnamon • unhealthy rats given cinnamon had fairly constant levels of glucose (D) • cinnamon lowered blood glucose by moving it into the liver (E) • cinnamon stimulated enzymes to convert glucose into glycogen (E) • so that it could not diffuse back into the blood (E) • levels of glucose were higher than in the healthy rats (D) • because cinnamon not as effective as insulin (CE) • possibly {not absorbed well / worked slowly / did not stimulate so many enzymes} (CE) • fluctuations in levels of glucose (D) • depending on how much glucose is used (E)	(6)

			Additional guidance
Level 0	0	No awardable content	
Level 1	1-2	An explanation may be attempted but with limited interpretation or analysis of the scientific information and with a focus on mainly just one piece of scientific information. The explanation will contain basic information, with some attempt made to link knowledge and understanding to the given context.	Simple descriptions of data with no real attempt of an explanation 1 mark = some description of the data 2 marks = detailed description of the data
Level 2	3-4	An explanation will be given, with occasional evidence of analysis, interpretation and/or evaluation of both pieces of scientific information. The explanation shows some linkages and lines of scientific reasoning, with some structure.	Some explanation given using the information given / own knowledge (E) 3 marks = simple explanation of what {cinnamon / insulin} is doing 4 marks = more detailed explanation of how {cinnamon / insulin} decreases glucose level
Level 3	5-6	An explanation is made that is supported throughout by sustained application of relevant evidence of analysis, interpretation and/or evaluation of both pieces of scientific information. The explanation shows a well-developed and sustained line of scientific reasoning, which is clear and logically structured.	Data logically explained with extended use of information provided (CE) 5 marks = plus a comparison made of effectiveness of cinnamon compared to insulin 6 marks = data explained in detail